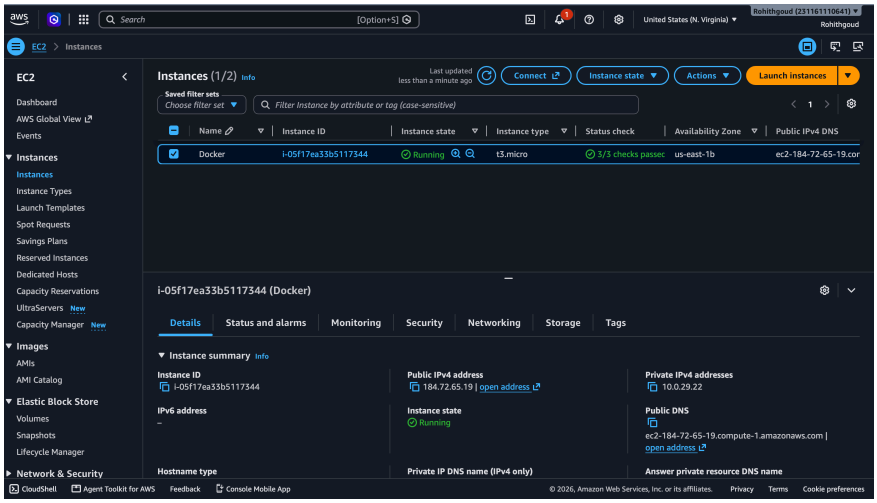
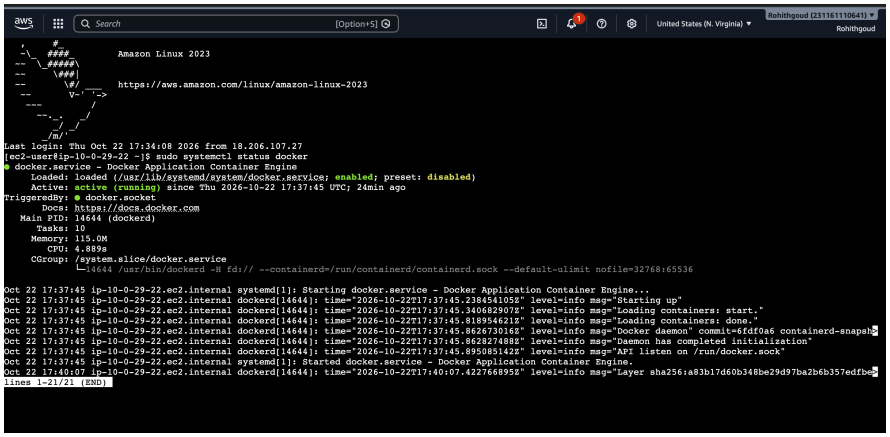
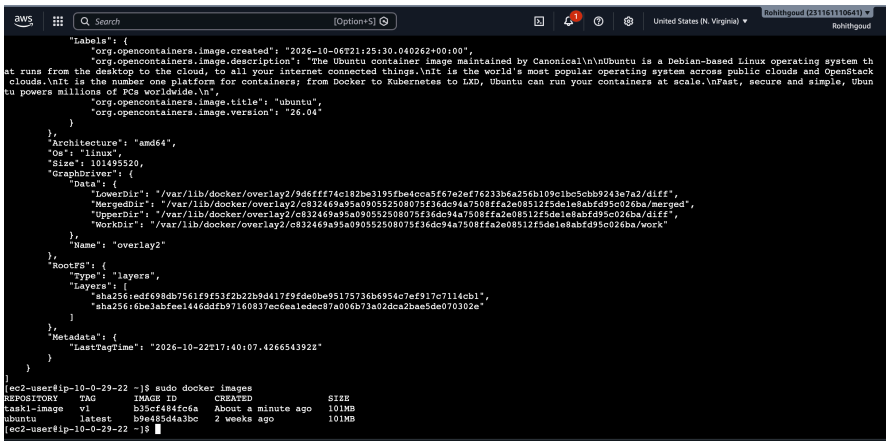
Docker-Task-3

1. Create a image from running container.

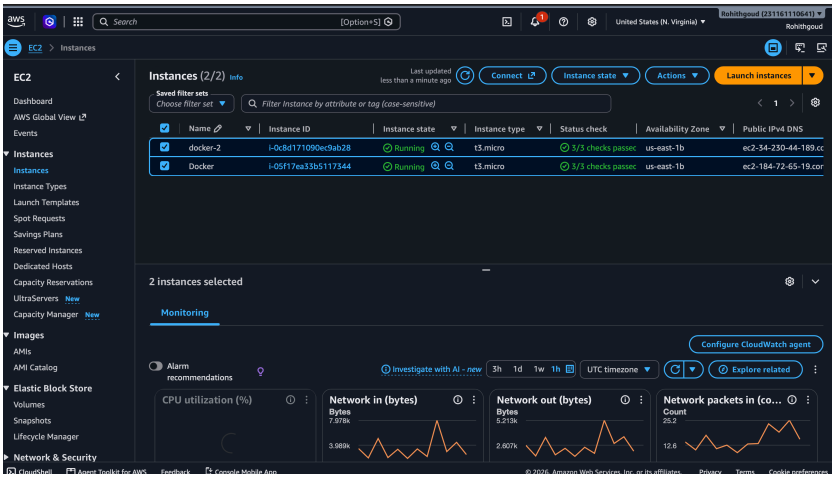






Docker-Task-3

2. Copy image from local machine to docker server and load the image.




```
PS C:\WINDOWS\system32> docker pull nginx:latest
latest: Pulling from library/nginx
113d9139ada2: Pull complete
f92a76c59dbd: Pull complete
c9ec797c69ed: Pull complete
cc510c1e359: Pull complete
188e9077f350: Pull complete
19352c60515: Pull complete
8a6757ab48b1: Pull complete
a5a8894bf2b2: Download complete
2b7d75aa4652: Download complete
Digest: sha256:332d0beea97cff8c0d5d27976c0778a4b00345b9d74c2feaaf8efde548314948
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
PS C:\WINDOWS\system32> docker images
                                                                    Info →  U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   11a3eb1597e8       25.9kB         9.49kB    U
nginx:latest         332d0beea97c        243MB         66.8MB
PS C:\WINDOWS\system32> docker save -o nginx.tar nginx:latest
PS C:\WINDOWS\system32> Get-Item .\nginx.tar


    Directory: C:\WINDOWS\system32


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        15-10-2026     17:45       66781696 nginx.tar
```

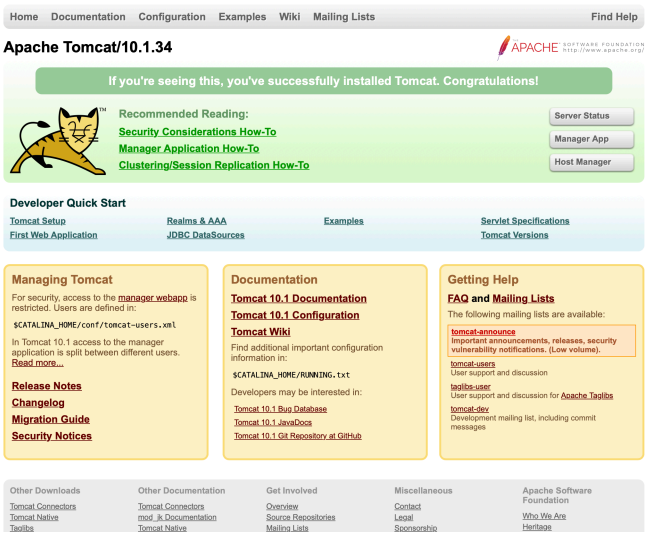
```
[root@ip-172-31-30-23 ~]# cd /tmp
[root@ip-172-31-30-23 tmp]# ls
nginx.tar  systemd-private-f634125Oe798427593ecfef6490b5305-chronyd.service-UDXhJJ  systemd-private-f634125Oe798427593ecfef6490b5305-policy-routes@ens5.service-sQ8EtM
systemd-private-f634125Oe798427593ecfef6490b5305-dbus-broker.service-oorQNH  systemd-private-f634125Oe798427593ecfef6490b5305-systemd-logind.service-Jz63U7
systemd-private-f634125Oe798427593ecfef6490b5305-systemd-resolved.service-3SubYj
[root@ip-172-31-30-23 tmp]# sudo docker load -i /tmp/nginx.tar
sudo docker images
Loaded image: nginx:latest
REPOSITORY   TAG       IMAGE ID       CREATED       SIZE
nginx        latest    ca5a66f340c7   18 hours ago  163MB
[root@ip-172-31-30-23 tmp]# sudo docker run -d --name nginx-server -p 8080:80 nginx:latest
d926ce0991a0894389d82baca03c98eae8ea99aacee12e8244dd4d6bdf80ba31
[root@ip-172-31-30-23 tmp]# sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS         PORTS                                     NAMES
d926ce0991a0   nginx:latest   "/docker-entrypoint.…"   17 seconds ago  Up 16 seconds  0.0.0.0:8080->80/tcp, :::8080->80/tcp     nginx-server
[root@ip-172-31-30-23 tmp]#
```

3. Create Docker image using alpine and customize with tomcat.

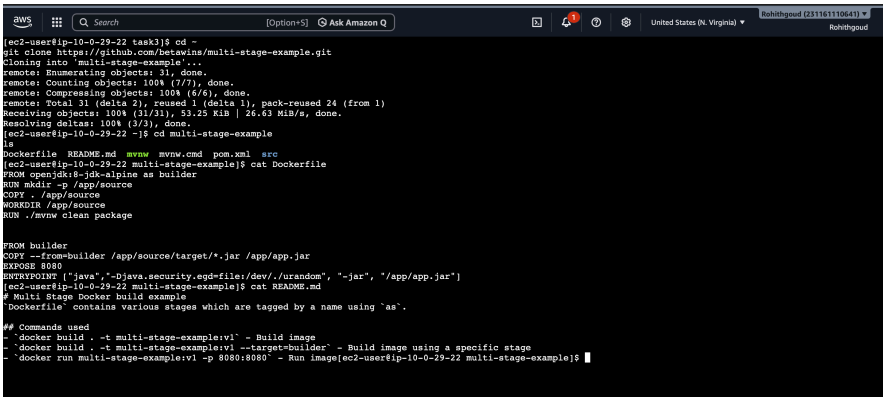


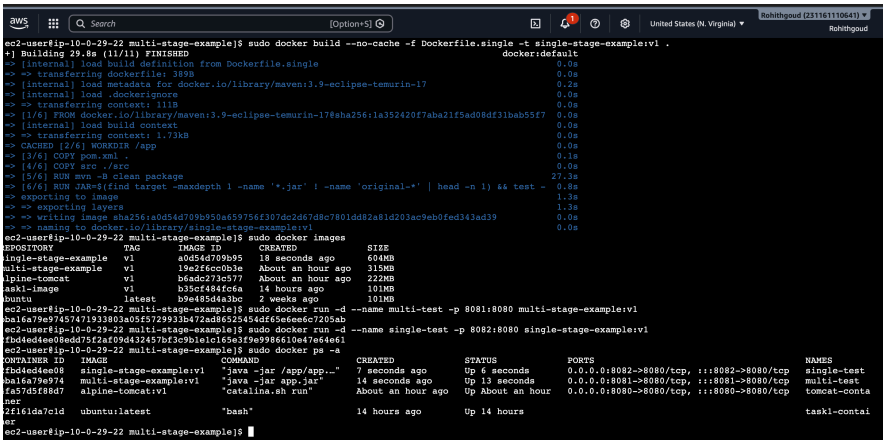
```
=> => transferring dockerfile: 584B                                                                    0.0s
=> [internal] load metadata for docker.io/library/alpine:3.20                                          0.5s
=> [internal] load .dockerignore                                                                       0.0s
=> => transferring context: 2B                                                                         0.0s
=> [1/2] FROM docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be121   0.3s
=> => resolve docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be121    0.0s
=> => extracting sha256:25f1d6b1951ac8eb3740558fe94cb83d377bdadf95fd8f98b50d2e1b96130471               0.1s
=> => sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be121613d44e6e2a4b6bc 9.23kB / 9.23kB         0.0s
=> => sha256:ce64c687cbea9300178b30c95835354e34c4e4febc4badfe27102879de0483b5e 1.02kB / 1.02kB         0.0s
=> => sha256:bf8527eb54e3680e728d5eb383a8ba730d72dae7216fbc8dff97ed6b224a731 612B / 612B               0.0s
=> => sha256:25f1d6b1951ac8eb3740558fe94cb83d377bdadf95fd8f98b50d2e1b96130471 3.63MB / 3.63MB          0.1s
=> [2/2] RUN apk add --no-cache openjdk17-jre curl     && mkdir -p /opt/tomcat     && curl -fSL htt  35.0s
=> exporting to image                                                                                  1.4s
=> => exporting layers                                                                                 1.4s
=> => writing image sha256:b6adc273c5778a17d1825175216d03b494f73d7af1fd8db224fee91cf517a231            0.0s
=> => naming to docker.io/library/alpine-tomcat:v1                                                     0.0s
[ec2-user@ip-10-0-29-22 task3]$ sudo docker images
REPOSITORY      TAG       IMAGE ID       CREATED          SIZE
alpine-tomcat   v1        b6adc273c577   39 seconds ago   222MB
task1-image     v1        b35cf484fc6a   12 hours ago     101MB
ubuntu          latest    b9e485d4a3bc   2 weeks ago      101MB
[ec2-user@ip-10-0-29-22 task3]$ sudo docker run -d --name tomcat-container -p 8080:8080 alpine-tomcat:v1
4fa57d5f88d7f38d3822ddd5e203f645472b87a1cb3c2dfe5da09a178c877bfe3
[ec2-user@ip-10-0-29-22 task3]$ sudo docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS          PORTS                                                  NAMES
4fa57d5f88d7   alpine-tomcat:v1     "catalina.sh run"        10 seconds ago   Up 8 seconds    0.0.0.0:8080->8080/tcp, :::8080->8080/tcp              tomcat-container
62f161da7c1d   ubuntu:latest        "bash"                   12 hours ago     Up 12 hours                                                            task1-container
[ec2-user@ip-10-0-29-22 task3]$ sudo docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED              STATUS              PORTS                                              NAMES
4fa57d5f88d7   alpine-tomcat:v1     "catalina.sh run"        About a minute ago   Up About a minute   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp          tomcat-container
62f161da7c1d   ubuntu:latest        "bash"                   12 hours ago         Up 12 hours                                                            task1-container
[ec2-user@ip-10-0-29-22 task3]$ sudo docker images
REPOSITORY      TAG       IMAGE ID       CREATED          SIZE
alpine-tomcat   v1        b6adc273c577   2 minutes ago    222MB
task1-image     v1        b35cf484fc6a   12 hours ago     101MB
ubuntu          latest    b9e485d4a3bc   2 weeks ago      101MB
[ec2-user@ip-10-0-29-22 task3]$ sudo docker inspect tomcat-container --format='{{.Config.Image}}'
alpine-tomcat:v1
[ec2-user@ip-10-0-29-22 task3]$
```

Docker-Task-3

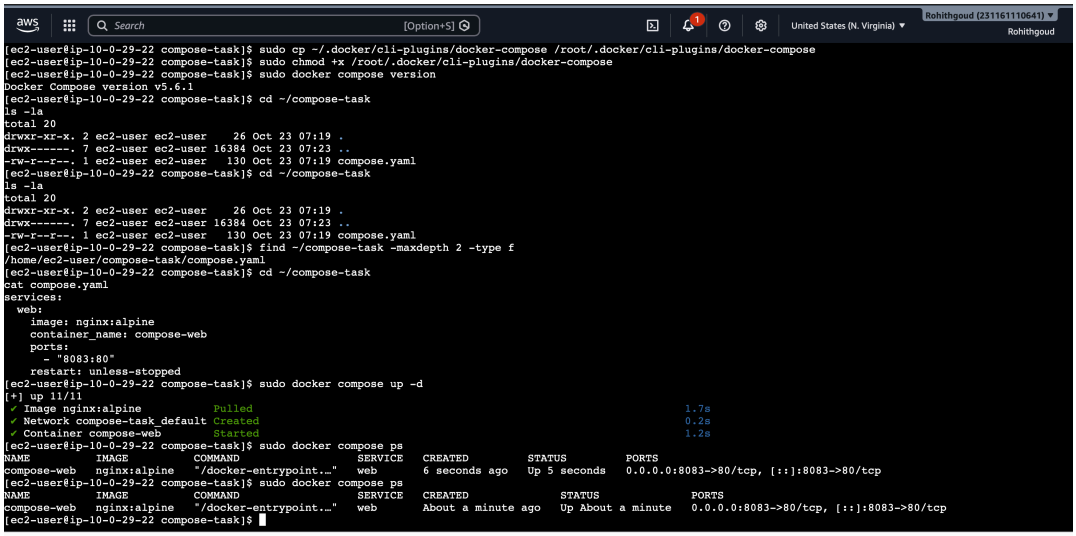


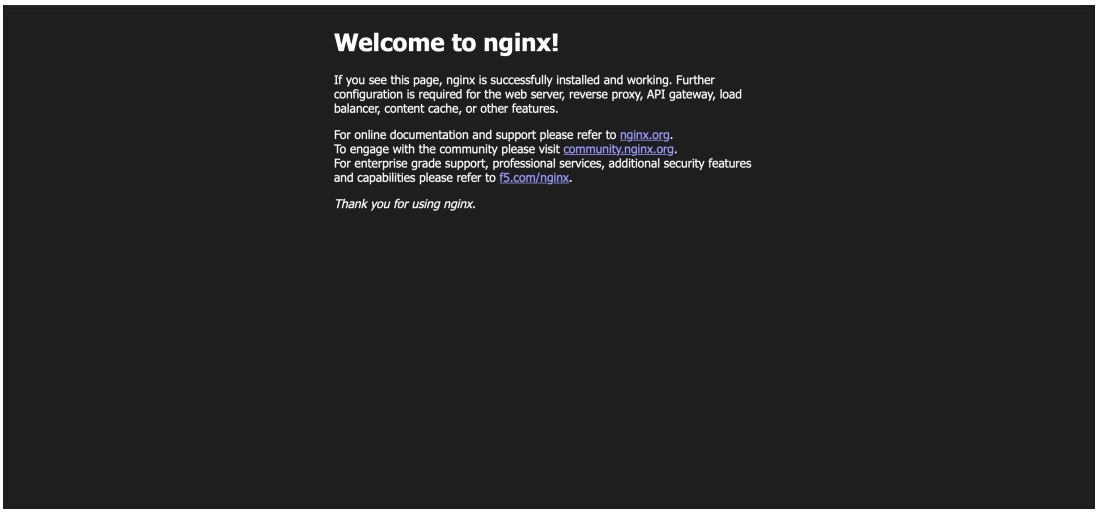
4. Create single stage and multi stage docker file using this source code: https://github.com/betawins/multi-stage-example.git



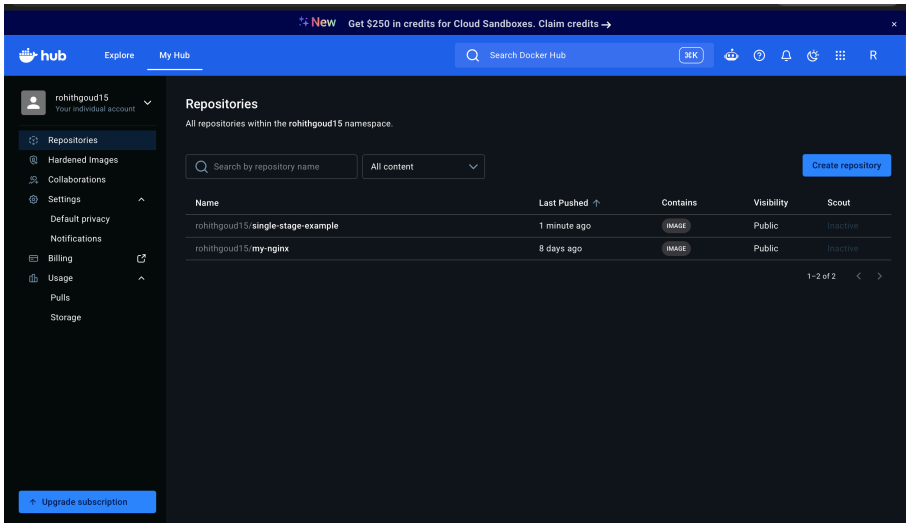


5. Install docker compose and execute sample application.



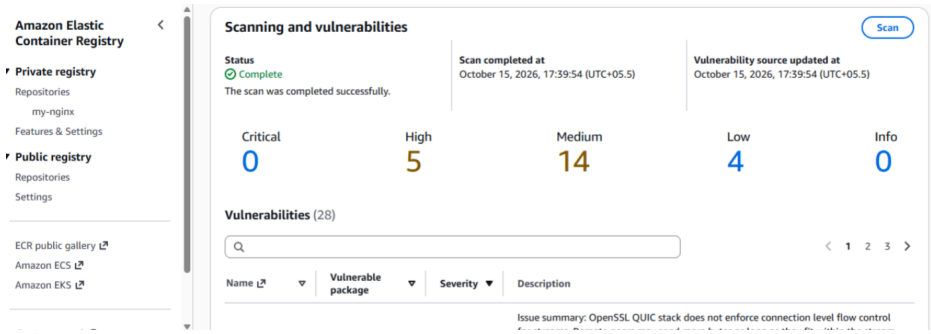


6. Implement solution to scan images when pushed to docker registry.



```
lpine: Pulling from library/nginx
igest: sha256:72aeb3308ef6cb3118242fb1638951892391 6dfa206e32d1a8377acc328b9870
tatus: Image is up to date for nginx:alpine
ocker.io/library/nginx:alpine
root@ip-172-31-13-222 ~]# docker images
EPOSITORY    TAG       IMAGE ID       CREATED        SIZE
y_image      1.0       1d49be0a8e04   2 hours ago    235MB
ginx         alpine    76090a34b858   17 hours ago   69.1MB
edis         alpine    15f5a4aad4c3   3 weeks ago    119MB
root@ip-172-31-13-222 ~]# docker tag nginx:alphine zeeshancloud15/zeeshan-horizon01
rror response from daemon: No such image: nginx:alphine
root@ip-172-31-13-222 ~]# docker tag nginx:alpine zeeshancloud15/zeeshan-horizon01
root@ip-172-31-13-222 ~]# docker push zeeshancloud15/zeeshan-horizon01
sing default tag: latest
he push refers to repository [docker.io/zeeshancloud15/zeeshan-horizon01]
aef7713033e: Mounted from library/nginx
5237c8bfc43: Mounted from library/nginx
cbec789df91: Mounted from library/nginx
01de5b74a88: Mounted from library/nginx
4b56df1052e: Mounted from library/nginx
1c068032a92: Mounted from library/nginx
d5f8e4b80b9: Mounted from library/nginx
4d97c428c51: Mounted from library/nginx
atest: digest: sha256:b09098d08f1d6da305f97808c3508a1924610f04a47c2e188999dd9c21037c08 size: 1989
root@ip-172-31-13-222 ~]#
```

7. Implement solution to scan images when pushed to AWS ECR.




```
  "registryId": "382736933724",
  "repositoryName": "my-nginx",
  "imageScanningConfiguration": {
      "scanOnPush": true
  }

oot@ip-172-31-13-222 ~]# docker push 382736933724.dkr.ecr.ap-south-1.amazonaws.com/my-nginx:1.0
e push refers to repository [382736933724.dkr.ecr.ap-south-1.amazonaws.com/my-nginx]
 image does not exist locally with the tag: 382736933724.dkr.ecr.ap-south-1.amazonaws.com/my-nginx
oot@ip-172-31-13-222 ~]# docker images
POSITORY                          TAG       IMAGE ID       CREATED        SIZE
 image                            1.0       1d49be0a8e04   3 hours ago    235MB
inx                               alpine    76090a34b858   17 hours ago   69.1MB
eshancloud15/zeeshan-horizon01    latest    76090a34b858   17 hours ago   69.1MB
dis                               alpine    15f5a4aad4c3   3 weeks ago    119MB
oot@ip-172-31-13-222 ~]# docker tag my_image:1.0 382736933724.dkr.ecr.ap-south-1.amazonaws.com/my-nginx:1.0
oot@ip-172-31-13-222 ~]# docker push 382736933724.dkr.ecr.ap-south-1.amazonaws.com/my-nginx:1.0
e push refers to repository [382736933724.dkr.ecr.ap-south-1.amazonaws.com/my-nginx]
4dc21b18d8: Preparing
ec93b572b7: Preparing
0a7d024626: Preparing
ab5dcca5fc: Preparing
 basic auth credentials
oot@ip-172-31-13-222 ~]#
```

8. Create a Jenkins pipeline to create a docker image and push the image to Docker hub.

